Chapter 2

2. Inference in Gaussian models.

- 2.1. MLE of μ and Σ.
- 2.2. Hotelling's test (Σ known).
- 2.3. Hotelling's test (Σ unknown).
- 2.4. Confidence zones.
- 2.4. Two-sample problems.

MLE of μ and Σ

Let $X_1, \ldots, X_n$ i.i.d. $\mathcal{N}_p(\mu, \Sigma)$. Then the MLE of μ and Σ are given as

$$
\begin{aligned}
(\hat{\mu}, \hat{\Sigma}) &= \operatorname{argmax}_{\mu \in \mathbb{R}^p, \Sigma > 0} L(\mu, \Sigma : X_1, \ldots, X_n) \\
&= \operatorname{argmax}_{\mu \in \mathbb{R}^p, \Sigma > 0} \prod_{i=1}^{n} \left[\frac{1}{|\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(X_i - \mu)' \Sigma^{-1} (X_i - \mu) \right) \right].
\end{aligned}
$$

The solution of this maximisation problem is well expected, and can be shown to be

$$
\hat{\mu} = \bar{X} := \frac{1}{n} \sum_{i=1}^{n} X_i
$$

and

$$
\hat{\Sigma} := \frac{1}{n} W := \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})(X_i - \bar{X})'.
$$

Hotelling's test (Σ known)

Let $X_1, \ldots, X_n$ i.i.d. $\mathcal{N}_p(\mu, \Sigma)$.

Let μ_0 be a fixed p-vector. A very important testing problem is

$$\begin{cases} \mathcal{H}_0 : \mu = \mu_0 \\ \mathcal{H}_1 : \mu \neq \mu_0. \end{cases}$$

For $p = 1$, it is natural (and common) to base the decision on $\bar{X}$ (more specifically on the distance between $\bar{X}$ and μ_0).

Hotelling's test (Σ known)

Since under $\mathcal{H}_0$ we have that $\bar{X} \sim \mathcal{N}_p(\mu_0, \frac{1}{n}\Sigma)$, it then follows that

$$T^2_{\text{mean}}(X) = n(\bar{X} - \mu_0)'\Sigma^{-1}(\bar{X} - \mu_0) = d^2_{\frac{1}{n}\Sigma}(\bar{X}, \mu_0) \sim \chi^2_p.$$

We reject $\mathcal{H}_0$ for large values of $T^2_{\text{mean}}(X)$. In particular:

We reject $\mathcal{H}_0$ at significance level α if $T^2_{\text{mean}}(X) > \chi^2_{p;1-\alpha}$.

Hotelling's test (Σ known)

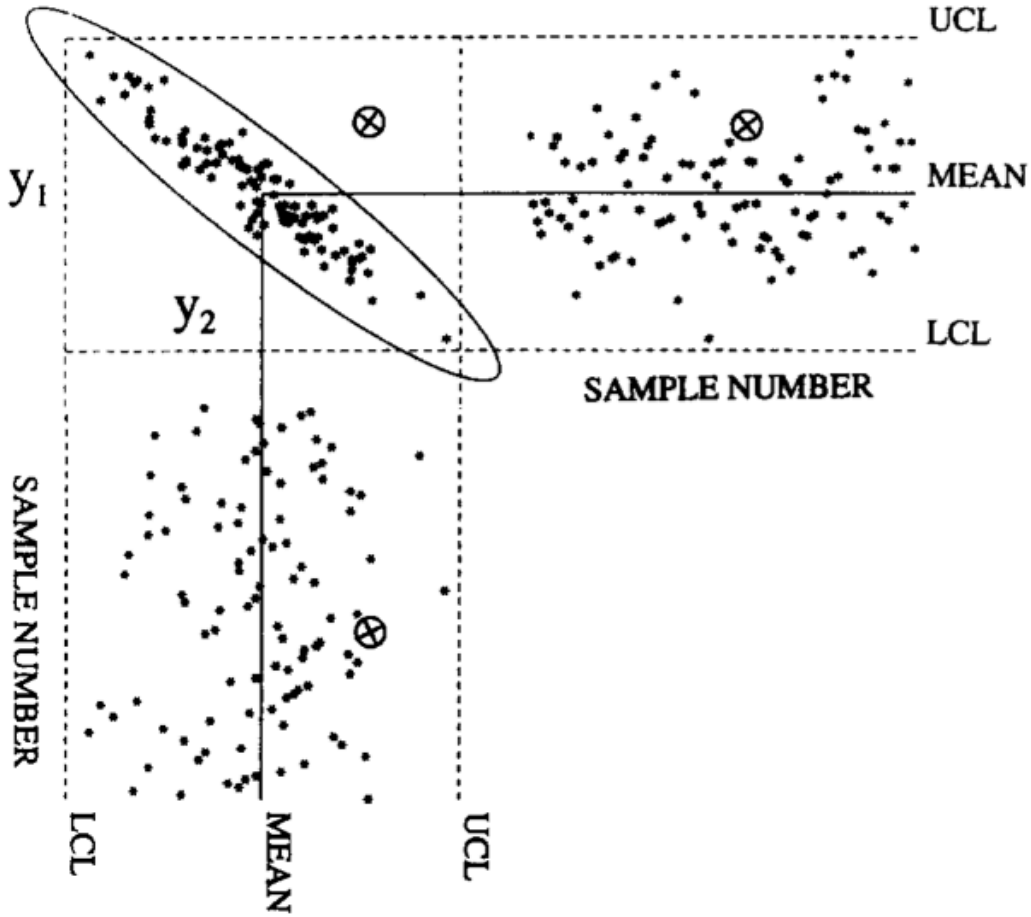
Example: Suppose we want to explore whether a production process/quality of water/etc. is in control or out of control. Suppose that in control measurements are p-variate (normal) data, which are iid (μ_0, Σ). The process is out of control if μ_0 changes.

On a daily (weekly, etc) basis we sample $n \geq 1$ data. For the i-th sample compute $T_i = T^2_{\text{mean}}(i\text{-th sample})$ and consider the chart $\{(i, T_i) : 1 \leq i \leq N\}$.

Check when (or if) the line is above $\chi^2_{p;0.99}$
$\Longleftrightarrow$ observation is outside 99% tolerance zone.

On the next slide, we show an illustrative example of a bivariate sample from MacGregor and Kourti (1995). *Statistical process control of multivariate processes*. The important point is the multivariate approach.

Hotelling's test (Σ known)



Hotelling's test (Σ known)

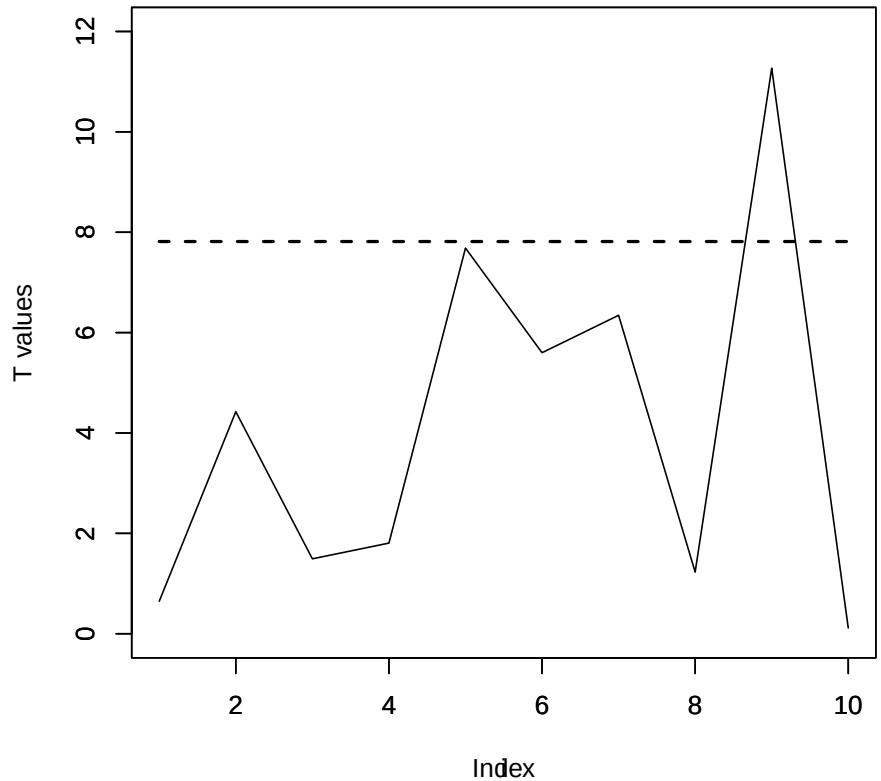


Figure 1: Simulated data: $\mu_0 = 0$, $n = 10$ (i.e. we compute 10 times the mean of 10 multivariate normals), $p = 3$, Σ known. Dashed line indicates 0.95 quantile of a χ^2_3-variable. Graph shows $10\bar{x}_i'\Sigma^{-1}\bar{x}_i$, $i = 1, \ldots, 10$. Our process was in control and we had one "false alarm".

Hotelling's test (Σ unknown)

If Σ is unknown it is natural to replace it by $\hat{\Sigma}$ or S.

$\rightsquigarrow T^2(X) = n(\bar{X} - \mu_0)' S^{-1}(\bar{X} - \mu_0)$ (usual notation: T^2).

When n is large, we can use that

$$T^2 \xrightarrow{d} \chi^2_p.$$

Hence, asymptotically (with $n \to \infty$) the level α-test again rejects $\mathcal{H}_0$ if $T^2 > \chi^2_{p;1-\alpha}$.

By the multivariate CLT this is still true if the $X_k \overset{\text{i.i.d.}}{\sim} (\mu, \Sigma)$ (no normal distribution required).

Hotelling's test (Σ unknown)

For $p = 1$

$$T^2 = \left| \frac{\sqrt{n}\,(\bar{X} - \mu_0)}{s} \right|^2, \text{ where } s^2 := \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2,$$

which is nothing but the square of Student's t-test.

If $X_1, \ldots, X_n$ are i.i.d. $\mathcal{N}_1(\mu_0, \sigma^2)$, then Fisher's lemma implies that

$$\frac{\sqrt{n}(\bar{X} - \mu_0)}{s} \sim t_{n-1}.$$

So the exact law of T^2 under $\mathcal{H}_0$ (for $p = 1$) is $F_{1,n-1}$.

The exact level-α-test rejects $\mathcal{H}_0$ if $T^2 > F_{1,n-1;1-\alpha}$.

Note that for the exact test we require a normal random sample.

Hotelling's test (Σ unknown)

One can derive the law of Hotelling's T^2 statistic

$$T^2 = n(\bar{X} - \mu_0)' S^{-1} (\bar{X} - \mu_0),$$

under $\mathcal{H}_0$ if the X_i are i.i.d. and p-variate normal:

$$T^2 \sim \frac{p(n-1)}{n-p} F_{p,n-p}.$$

A level-α-test rejects $\mathcal{H}_0$ if

$$\frac{n-p}{p(n-1)} T^2 > F_{p,n-p;1-\alpha}.$$

Example

We consider the `iris` data set in `R`.

```
> head(iris)

Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa


# choose Species versicolor and remove the categorial parameter Species
> iris.vers=iris[which(iris$Species=="versicolor"),-5]
```

Example

```
# Package containing Hotelling's test
> install.packages("DescTools")
> library("DescTools")

#Nullhypothesis
> mu0=c(5.5,2.6,4.0,1.3)

# We test H0: mu=mu0 using F-Test
HotellingsT2Test(iris.vers,mu=mu0,test="f")

Hotelling's one sample T2-test

data:  iris.vers
T.2 = 12.543, df1 = 4, df2 = 46, p-value = 5.59e-07
alternative hypothesis: true location is not equal to c(5.5,2.6,4,1.3)
```

Here $T.2 = \frac{n-p}{p(n-1)} n(\bar{X} - \mu_0)' S^{-1} (\bar{X} - \mu_0)$, $df1 = p$, $df2 = n - p$.

Example

```
# We test H0: mu=mu0 using chi2-Test
> HotellingsT2Test(iris.vers,mu=mu0,test="chi")

Hotelling's one sample T2-test

data:  iris.vers
T.2 = 53.442, df = 4, p-value = 6.886e-11
alternative hypothesis: true location is not equal to c(5.5,2.6,4,1.3)
```

Here $T.2 = n(\bar{X} - \mu_0)' S^{-1} (\bar{X} - \mu_0)$, $df = p$.

Confidence zones

Definition

We call a (random set) $C_{1-\alpha}^{(n)} = C_{1-\alpha}(X_1, \ldots, X_n) \subset \mathbb{R}^p$ a $(1-\alpha) \times 100\%$ confidence zone for a parameter θ if

$$P_\theta(\theta \in C_{1-\alpha}^{(n)}) \geq 1 - \alpha.$$

For example, if $X_1, \ldots, X_n$ i.i.d. $\mathcal{N}_p(\mu_0, \Sigma)$, then we saw that

$$P\left[n \underbrace{(\bar{X} - \mu_0)' S^{-1} (\bar{X} - \mu_0)}_{d_S^2(\bar{X}, \mu_0)} \leq \frac{p(n-1)}{n-p} F_{p,n-p;1-\alpha} \right] = 1 - \alpha.$$

Confidence zones

Hence

$$C_{1-\alpha}^{(n)} := \left\{ \mu \in \mathbb{R}^p \,\middle|\, n d_S^2(\bar{X}, \mu) \leq \frac{p(n-1)}{(n-p)} F_{p,n-p;1-\alpha} \right\}$$

is a $(1-\alpha) \times 100\%$ confidence zone for μ_0.

Confidence zones

Similarly,

$$P\left[nd_S^2(\bar{X}, \mu_0) \leq \chi^2_{p;1-\alpha}\right] \to 1-\alpha, \quad (n \to \infty),$$

implies that an asymptotic $(1-\alpha) \times 100\%$ confidence zone is given by the ellipsoïde

$$C_{1-\alpha}^{(\infty)} := \left\{\mu \in \mathbb{R}^p \,\middle|\, nd_S^2(\bar{X}, \mu) \leq \chi^2_{p;1-\alpha}\right\}.$$

Remark: The latter result doesn't require normal data (only second order moments and sufficiently large n).

Example

Exact (red) and asymptotic (black) confidence ellipses for $X_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}_2(\mu, \Sigma)$, where $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$ et $\Sigma = \begin{pmatrix} 5 & 3 \\ 3 & 2.25 \end{pmatrix}$.

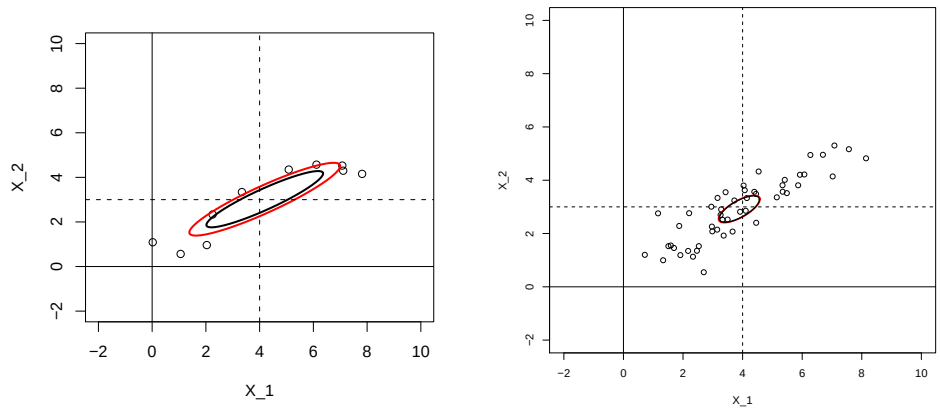


Figure 2: Left panel: $n = 10$. Right panel: $n = 50$.

Table 1: Proportion of μ not contained in the confidence zone for 10.000 M-C-simulations.

	$n = 10$	$n = 50$
exact	0.051	0.049
asymptotic	0.127	0.06

Confidence zones

Often it is prefered to have rectangular confidence zones:

$$P(L_1 \leq \mu_1 \leq U_1, \ldots, L_p \leq \mu_p \leq U_p) = 1 - \alpha.$$

If the components were independent, we could construct individual confidence intervals for each component

$$P(L_k \leq \mu_k \leq U_k) = 1 - \alpha_k$$

and such that $\prod_{k=1}^{p}(1 - \alpha_k) = 1 - \alpha$. Then by independence

$$P(L_1 \leq \mu_1 \leq U_1, \ldots, L_p \leq \mu_p \leq U_p) = 1 - \alpha.$$

1. For small values of α we have $(1 - \alpha/p)^p \approx 1 - \alpha$.
2. Rectangles (as all confidence zones) are not unique! (Choose $\alpha_1 = \alpha$ and $\alpha_k = 0$ for $k = 2, \ldots, p$.)

Confidence zones

Using the elementary fact that

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1$$

we can deduce that if $P(A_i) \geq 1 - \alpha_i$, then

$$P(A_1 \cap A_2 \cap \ldots \cap A_p) \geq 1 - (\alpha_1 + \alpha_2 + \cdots + \alpha_p).$$

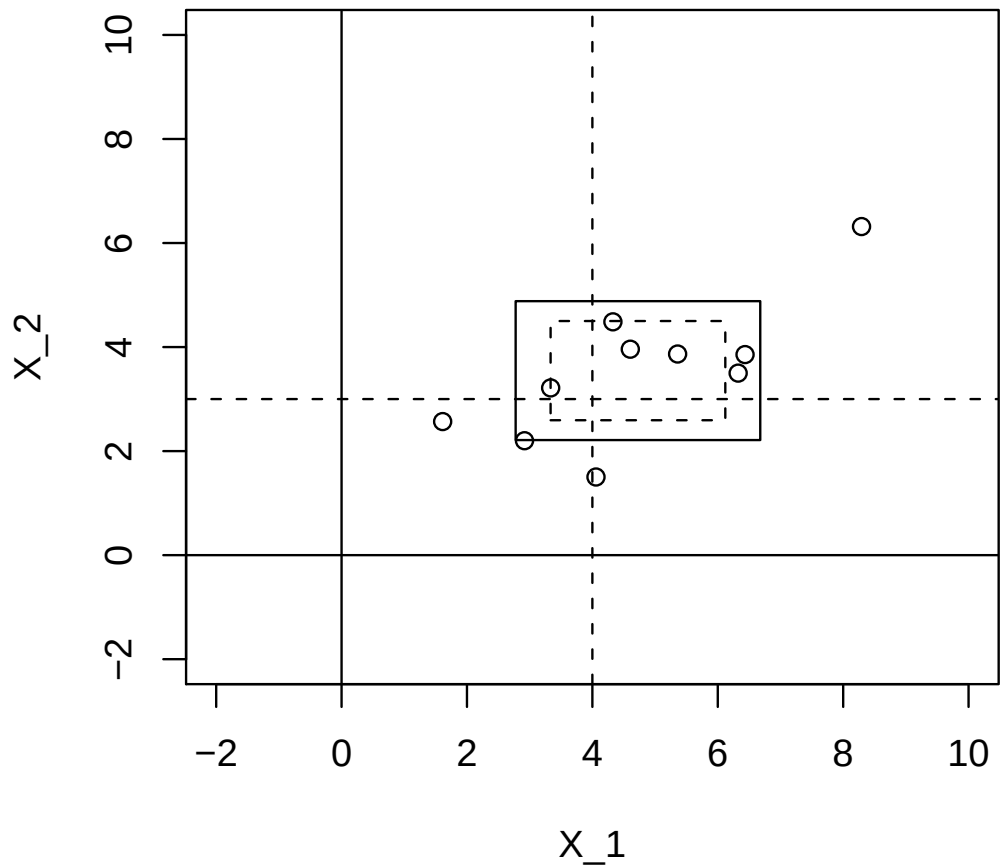
In particular choosing $\alpha_i = \alpha/p$ we obtain that

$$P(L_1 \leq \mu_1 \leq U_1, \ldots, L_p \leq \mu_p \leq U_p) \geq 1 - \alpha,$$

if the $[L_i, U_i]$ are individual $(1 - \alpha/p) \times 100\%$ confidence intervals. We call this approach the Bonferroni method.

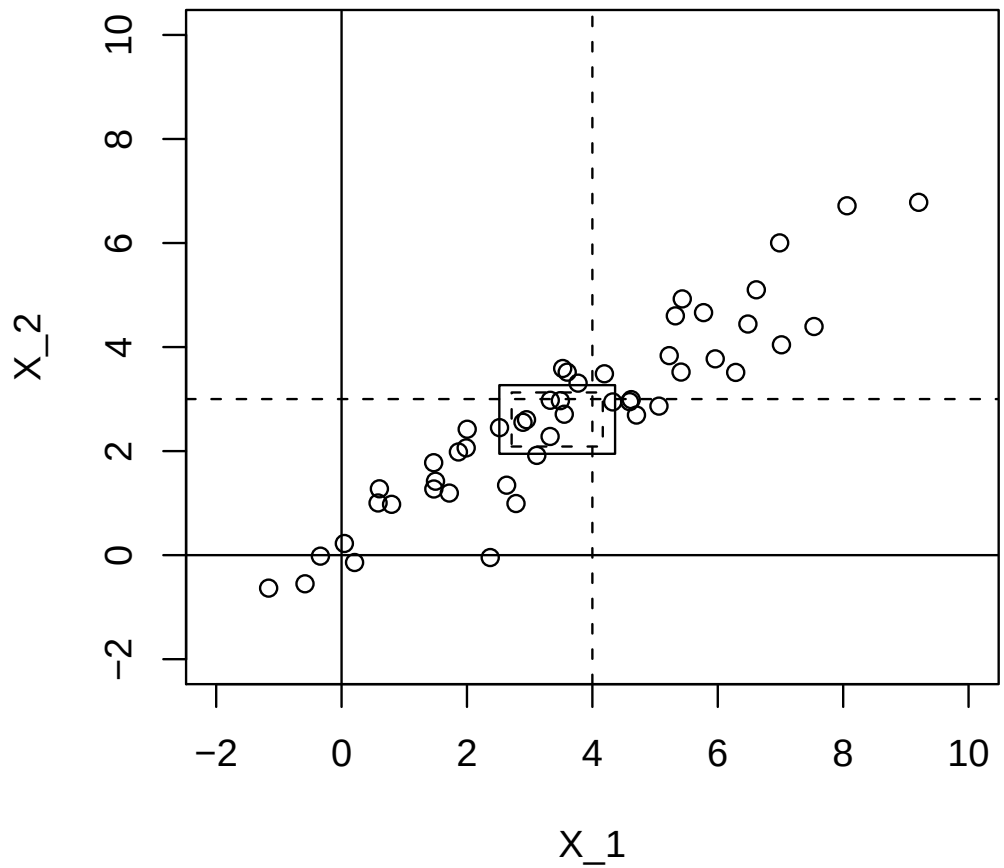
Confidence zones

Simultaneous (solid) v.s. individual (dashed) confidence zones for i.i.d. sample $\mathcal{N}_2(\mu, \Sigma)$ with $n = 10$, $\mu = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ et $\Sigma = \begin{pmatrix} 5 & 3 \\ 3 & 2.25 \end{pmatrix}$.



Confidence zones

Simultaneous (solid) v.s. individual (dashed) confidence zones for i.i.d. sample $\mathcal{N}_2(\mu, \Sigma)$ with $n = 50$, $\mu = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ et $\Sigma = \begin{pmatrix} 5 & 3 \\ 3 & 2.25 \end{pmatrix}$.



Example

We want a 95% rectangular confidence zone for the iris data from above.

We calculate $(1 - \frac{0.05}{4}) \times 100\%$ CI's for each component.

```
# compute standard deviations of components
> sigmas=sqrt(diag(cov(iris.vers)))
# compute sample mean vector
> hatmu=colMeans(iris.vers)
# compute t_{n-1}(1-0.05/4) quantiles
> tquant=qt(0.9875,49)
# lower bounds
> hatmu-tquant*sigmas/sqrt(50)

Sepal.Length  Sepal.Width Petal.Length  Petal.Width
    5.767202     2.667382     4.106330     1.261331

# upper bounds
> hatmu+tquant*sigmas/sqrt(50)

Sepal.Length  Sepal.Width Petal.Length  Petal.Width
    6.104798     2.872618     4.413670     1.390669
```

Two-sample problems

Consider two independent random samples $X_1, \ldots, X_{n_1}$ i.i.d. $\mathcal{N}_p(\mu_1, \Sigma)$ and $Y_1, \ldots, Y_{n_2}$ i.i.d. $\mathcal{N}_p(\mu_2, \Sigma)$.

We consider the testing problem

$$\begin{cases} \mathcal{H}_0 : \mu_1 = \mu_2 \\ \mathcal{H}_1 : \mu_1 \neq \mu_2. \end{cases}$$

Remark: More delicate is the problem with 2 different covariance matrices (Behrens-Fisher problem, Welch's t-test).

In our case, the tests will be based on $\bar{X} - \bar{Y}$.

Two-sample problems

We define

$$T^2 = \left(\frac{1}{n_1}+\frac{1}{n_2}\right)^{-1}(\bar{X}-\bar{Y})' S_{\text{pool}}^{-1}(\bar{X}-\bar{Y}) = \left(\frac{1}{n_1}+\frac{1}{n_2}\right)^{-1} d^2_{S_{\text{pool}}}(\bar{X},\bar{Y}),$$

where

$$\bar{X} := \frac{1}{n_1}\sum_{i=1}^{n_1} X_i, \;\; W_x := \sum_{i=1}^{n_1}(X_i-\bar{X})(X_i-\bar{X})',$$

$$\bar{Y} := \frac{1}{n_2}\sum_{i=1}^{n_2} Y_i, \;\; W_y := \sum_{i=1}^{n_2}(Y_i-\bar{Y})(Y_i-\bar{Y})',$$

and

$$S_{\text{pool}} := \frac{W_x + W_y}{n_1+n_2-2}.$$

$\mathcal{H}_0 : \mu_1 = \mu_2$ is rejected for large values of T^2.

Two-sample problems

Proposition

Suppose that $n_1 + n_2 \geq p + 2$. *Then under* $\mathcal{H}_0$,

$$\frac{n_1 + n_2 - p - 1}{p(n_1 + n_2 - 2)} T^2 \sim F_{p, n_1+n_2-p-1}.$$

Hotelling's exact test rejects

$$\mathcal{H}_0 \colon \mu_1 = \mu_2$$

at level α if

$$\frac{n_1 + n_2 - p - 1}{p(n_1 + n_2 - 2)} T^2 > F_{p, n_1+n_2-p-1; 1-\alpha}.$$

Asympotically $\mathcal{H}_0 \colon \mu_1 = \mu_2$ is rejected if $T^2 > \chi^2_{p; 1-\alpha}$.

Example

We compare the species `setosa` with `versicolor` in the `iris` data.

```
# Two sample test
> HotellingsT2Test(iris.vers,iris.seto,test="f")

Hotelling's two sample T2-test

data:  iris.vers and iris.seto
T.2 = 625.46, df1 = 4, df2 = 95, p-value < 2.2e-16
alternative hypothesis: true location difference is not equal to c(0,0,0,0)
```

Two-sample problems

We have assumed that $(X_1, \ldots, X_{n_1})$ and $(Y_1, \ldots, Y_{n_2})$ are independent.

In case of paired observations (e.g. systolic and diastolic blood pressure before and after some treatment) this may be very unrealistic.

Suppose that $(Z_1, \ldots, Z_n)$ is a random sample $\mathcal{N}_{2p}(\mu, \Sigma)$ with $Z_i = (X_i', Y_i')'$ and $X_i \sim \mathcal{N}_p(\mu_1, \Sigma_1)$ and $Y_i \sim \mathcal{N}_p(\mu_2, \Sigma_2)$.

We want to test $\mathcal{H}_0 \colon \mu_1 = \mu_2$. To this end we form the sample of differences $Y_1 - X_1, \ldots, Y_n - X_n$ and perform a test for $\mathcal{H}_0 \colon \mu = 0$.